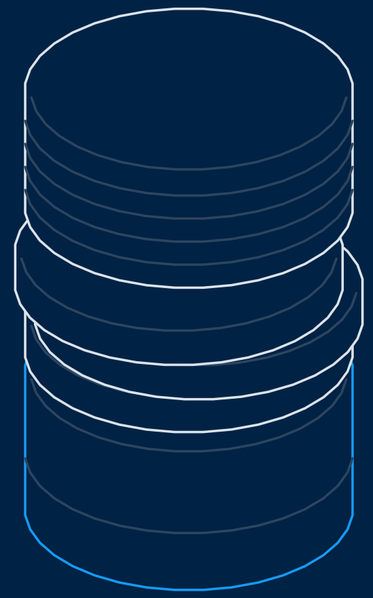
WORLD BANK GROUP

Strata

Procurement intelligence on World Bank open data, with every claim cited

Built for the World Bank Group's procurement intelligence assessment.



Chirag Lakhanpal

November 1, 2026

Strata is an independent demo built on World Bank open data. It is not a World Bank product, and the World Bank has not reviewed or endorsed it.

The problem, a live demo, then how Strata is built, answers and grows

1 **Three strata**
A procurement plan needs records, operational documents and evaluations, and today they sit apart.

2 **Strata at work**
Strata checks a borrower's plan, sizes a package it marks Watch and answers a question about it, live.

3 **How it is built**
World Bank open data and documents become one dated knowledge model on Databricks.

4 **How it answers**
The agent picks its tools, and the verifier checks every sentence before anyone sees it.

5 **How it grows**
Strata learns while people decide each change, and a pilot would grow in phases.

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A procurement plan needs three strata of evidence, and today they sit apart



Source: the assessment brief, section 1.

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Today the evidence reaches a plan too late to change it

- Teams set schedules without knowing how long comparable packages took.
- Packages are advertised again, or canceled, for reasons a past report already named.
- Specifications miss what the market can deliver.
- Lessons sit in PDF archives, so the same mistakes repeat.

> How does your team find what a completion report said about packages like this one?

Source: the assessment brief, section 1.

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The borrower owns the plan, and Strata helps both sides check it

World Bank staff

Procurement specialists and task team leaders review a borrower's plan and prepare missions. They leave with review notes and a mission brief.

Implementing agency officers

They draft the Procurement Plan and check it before they send it. They leave with a planned date for each step, and risks with their sources.

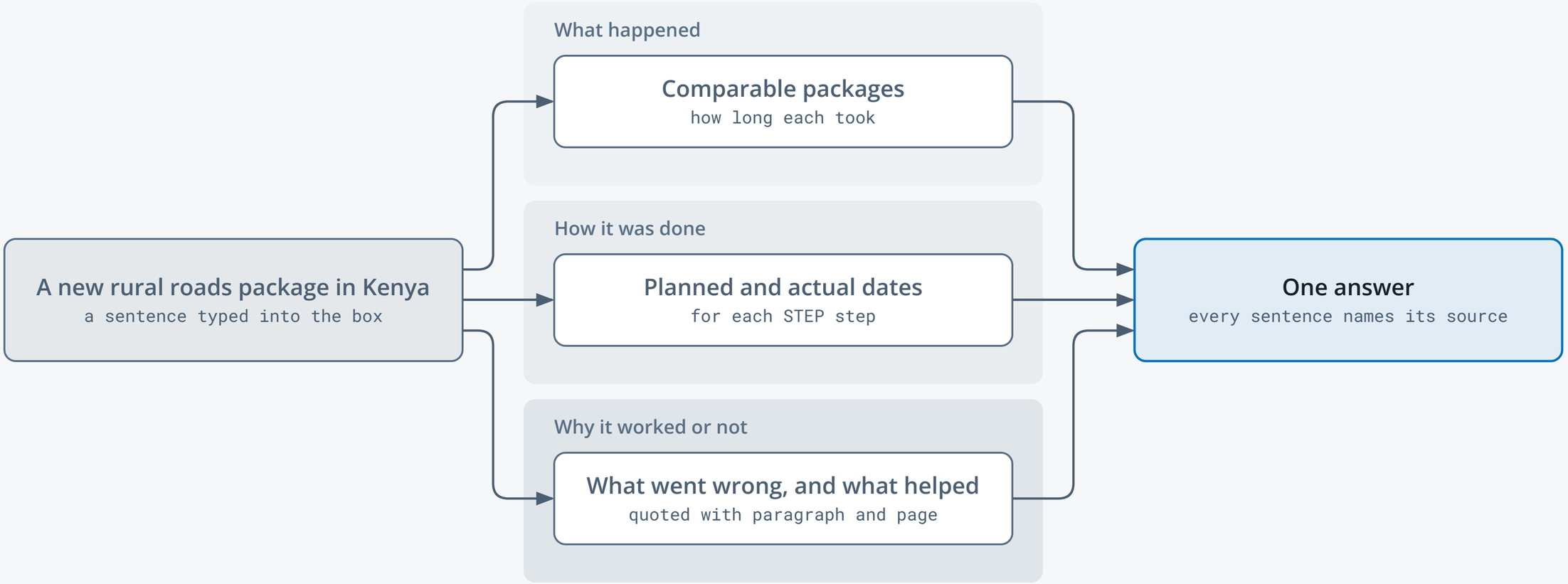
The public, read-only

They follow a contract from its notice to its completion report, in neutral words. Active projects show them no signals or estimates.

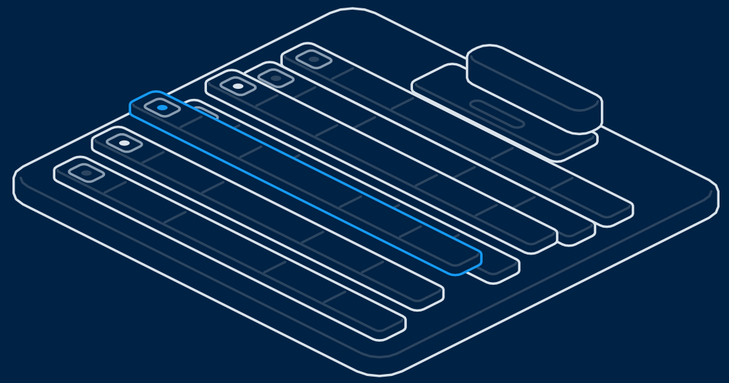
Strata drafts, and people decide. Exporting sends nothing, and nothing Strata writes is a World Bank decision.

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Strata joins the three strata once, so one question gets one cited answer



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2

Strata at work

Strata checks a borrower's plan, sizes a package it marks Watch and answers a question about it, live.

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Checking a borrower's Procurement Plan, row by row

Northern Water Works Development Agency's Procurement Plan of October 9, 2026, P156634

Open the plan check

- Strata checks each activity against past ranges, review thresholds and the plan's own dates.
- A signal is a reason to look, never a finding, and it names the record behind it.

Source: the Procurement Plan of October 9, 2026 for P156634, as published in Systematic Tracking of Exchanges in Procurement (STEP).

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Comparing that package with the packages that came before it

Package KE-NWWDA-541055-CW-RFB, which the plan check marks Watch

Show the recorded answer

Open the package

- Each comparable package is a dot on a month axis, from its notice to its signed contract.
- Hollow dots are still open, and the range counts them, so the slow cases stay in.
- The time estimate comes from a survival model. With no model or too few packages, it says why instead of guessing.

Note: STEP is Systematic Tracking of Exchanges in Procurement, where Procurement Plans are kept.

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Asking about the package, with every number checked against the records

How long did comparable packages take from notice to signed contract?

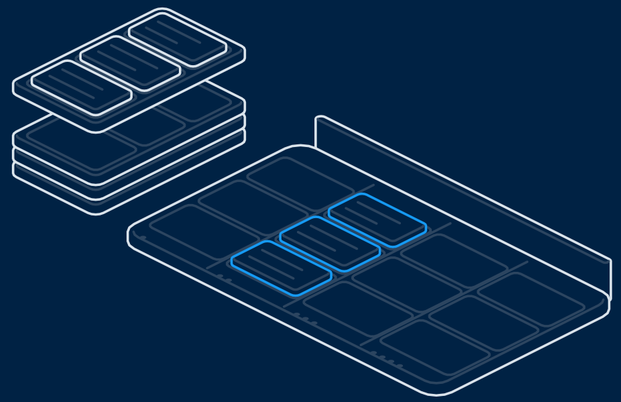
Copy the question

Open the package

- Every sentence names its source, and one click opens the record behind it.
- The verifier checks every number against the records before the answer shows.
- A sentence it cannot back is removed and shown, with the reason.

Note: the verifier runs on every answer and is never bypassed.

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3

How it is built

World Bank open data and documents become one dated knowledge model on Databricks.

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The data, search and tools run on Databricks, and the agent and its models run in Microsoft Foundry

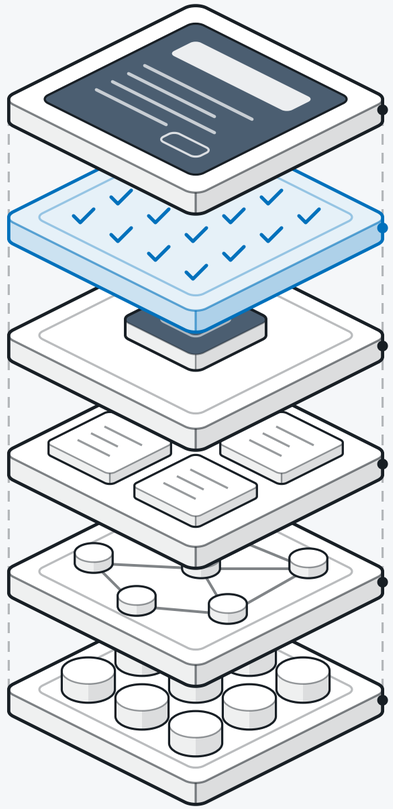
Where it runs

Databricks

Microsoft Foundry

● runs there

language models: Azure OpenAI, through Foundry



Strata app
a Databricks App:
runs each tool as the viewer

The verifier
checks every sentence
before anyone sees it

Strata agent
Microsoft Foundry: picks the tools,
writes claims with their evidence IDs

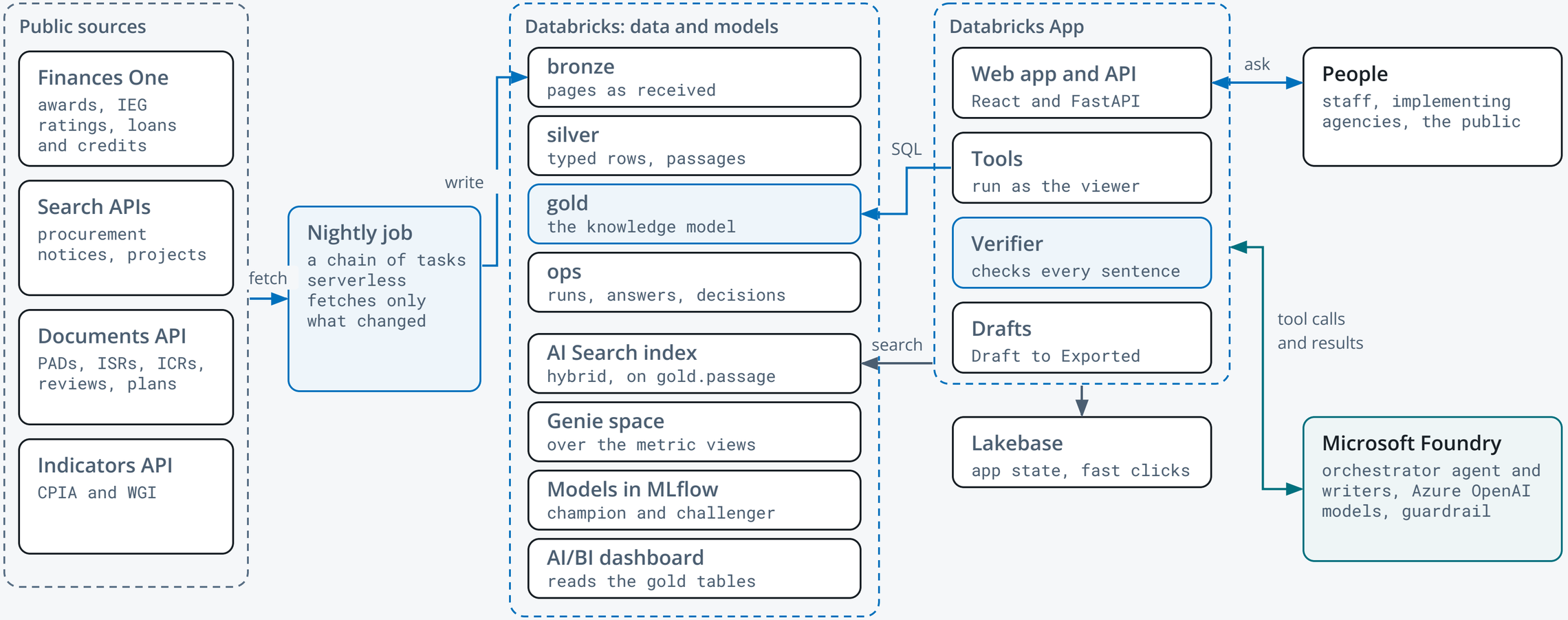
Tools
fixed SQL on the gold tables, hybrid search,
a Genie space, time estimates

Knowledge model
Unity Catalog tables, an edge table,
numbered passages, dated assertions

Data
World Bank open data, read every night
into bronze, silver and gold

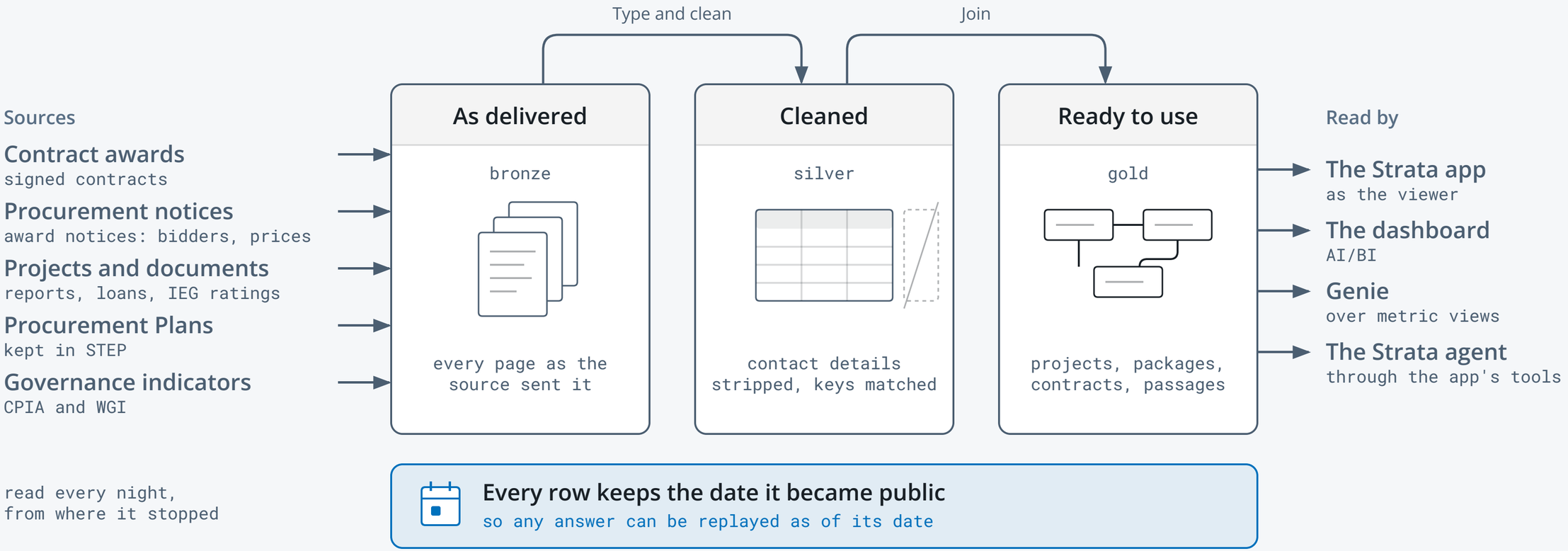
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The whole system, from the public sources to the person asking



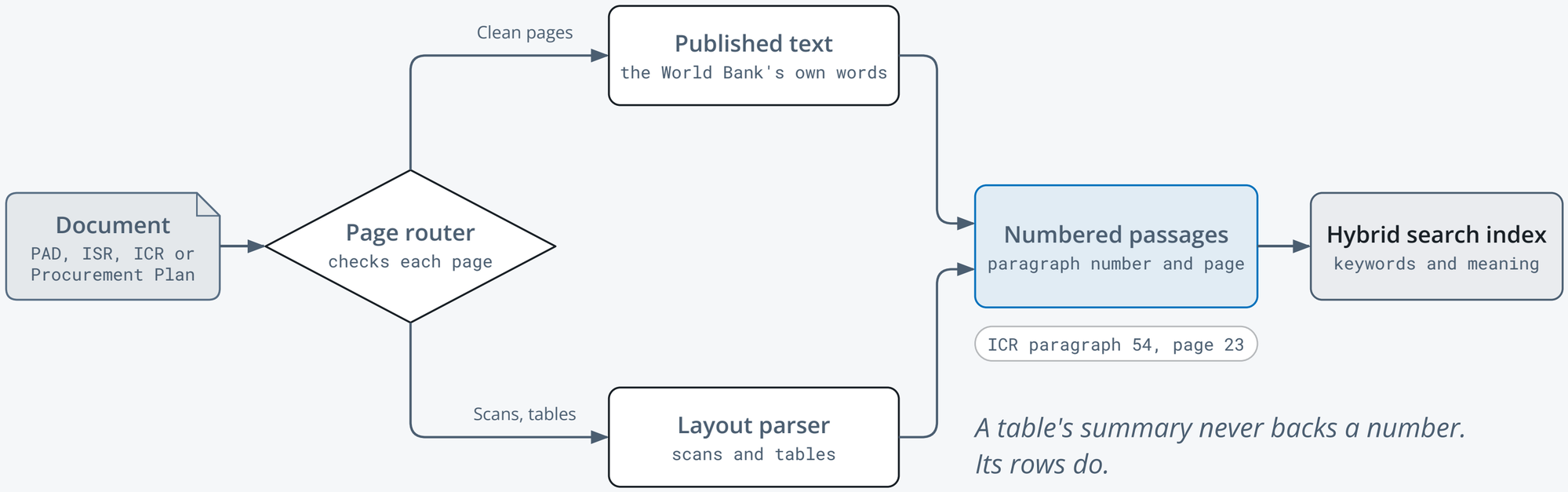
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Every night, World Bank open data become one dated knowledge model



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Every document becomes numbered passages, each with its paragraph and page



A table's summary never backs a number. Its rows do.

◌ Strata has not counted its passages or measured how well it read its pages yet.

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What the data looks like, and where it breaks

The awards datasets, and the gap

296,096 award rows since fiscal 2020 and 146,802 for 2001 to 2016. None cover 2017 to 2019, so Strata built 2,615 contracts from award notices.



The missing key

Only 8,059 of 283,867 contract award rows since fiscal 2020 share their STEP reference with a procurement notice.

Suppliers

4,957 of 5,488 awarded bidders give one name and one STEP supplier ID across their award rows. The rest need resolving.

Contact details

Ingestion removed contact details from 12,445 of 32,101 procurement notices, and the latest nightly check found 0 left.

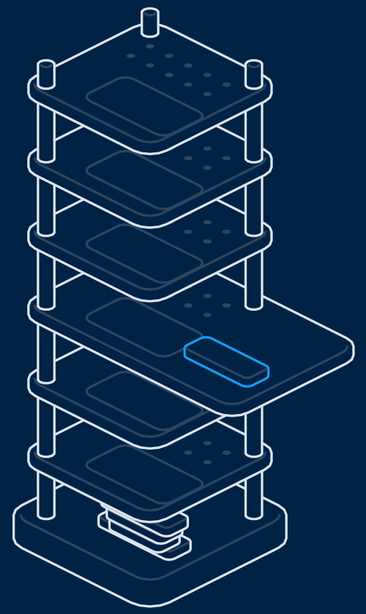
Reports

4,154 numbered passages, and 144 disagreements between reports and records.

Open the Data readiness page

Source: Strata's Data readiness measures, pulled November 1, 2026.

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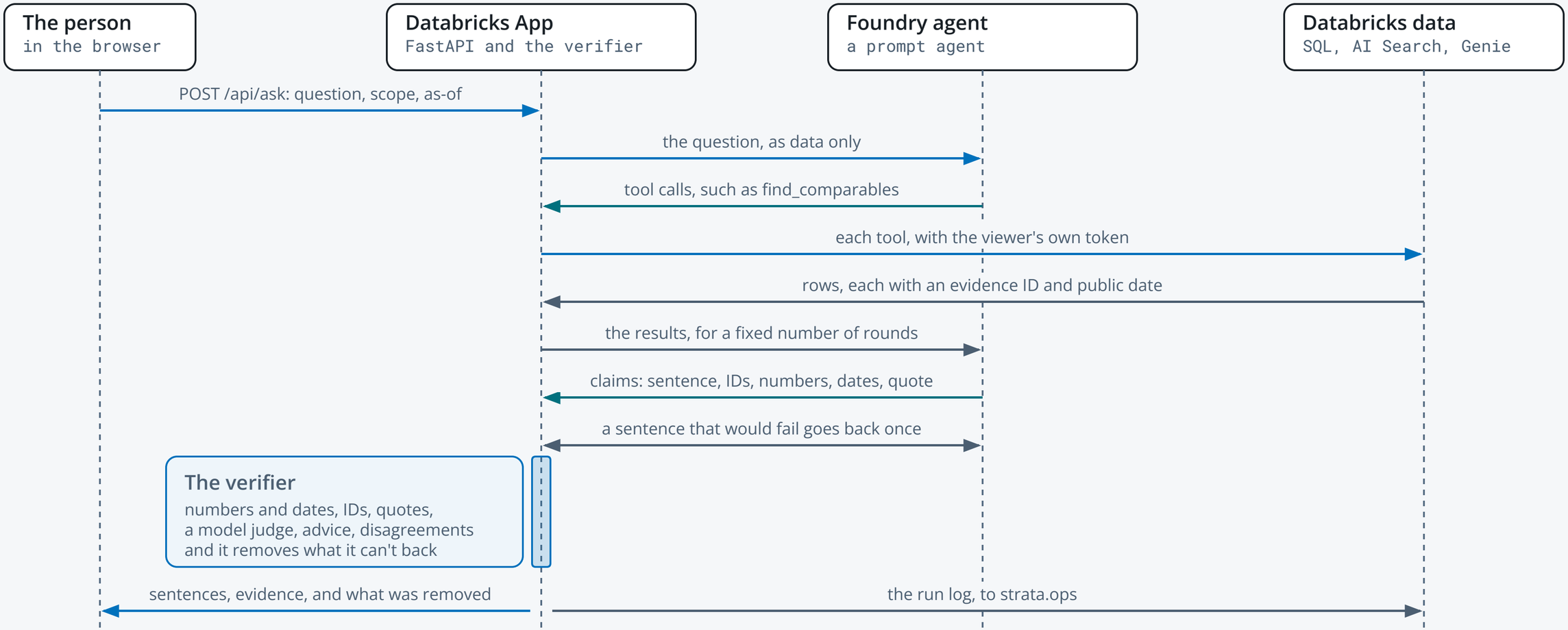
4

How it answers

The agent picks its tools, and the verifier checks every sentence before anyone sees it.

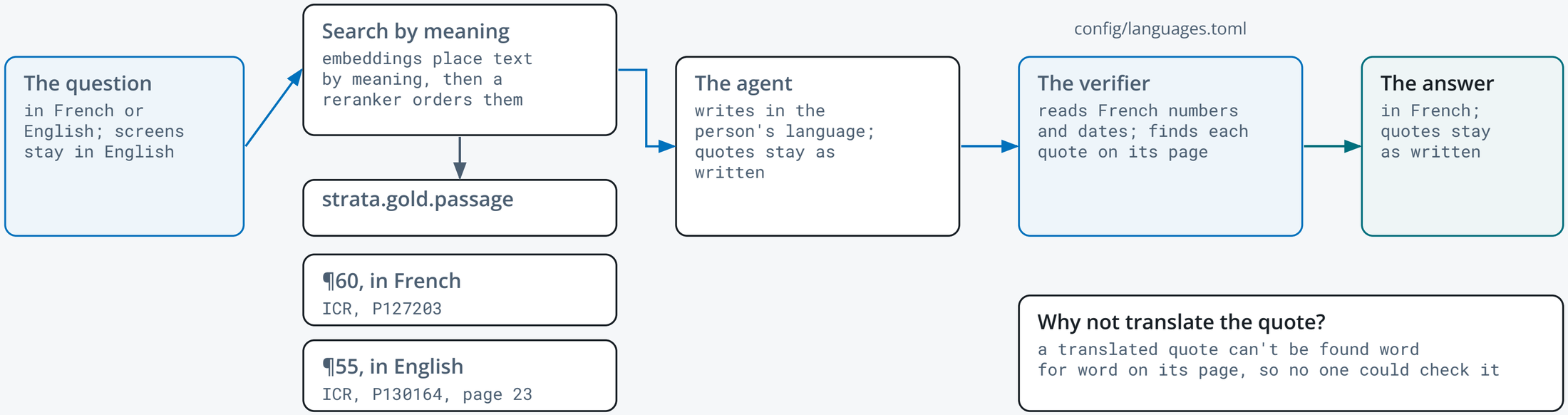
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The agent picks the tools, and the app runs them as the person asking



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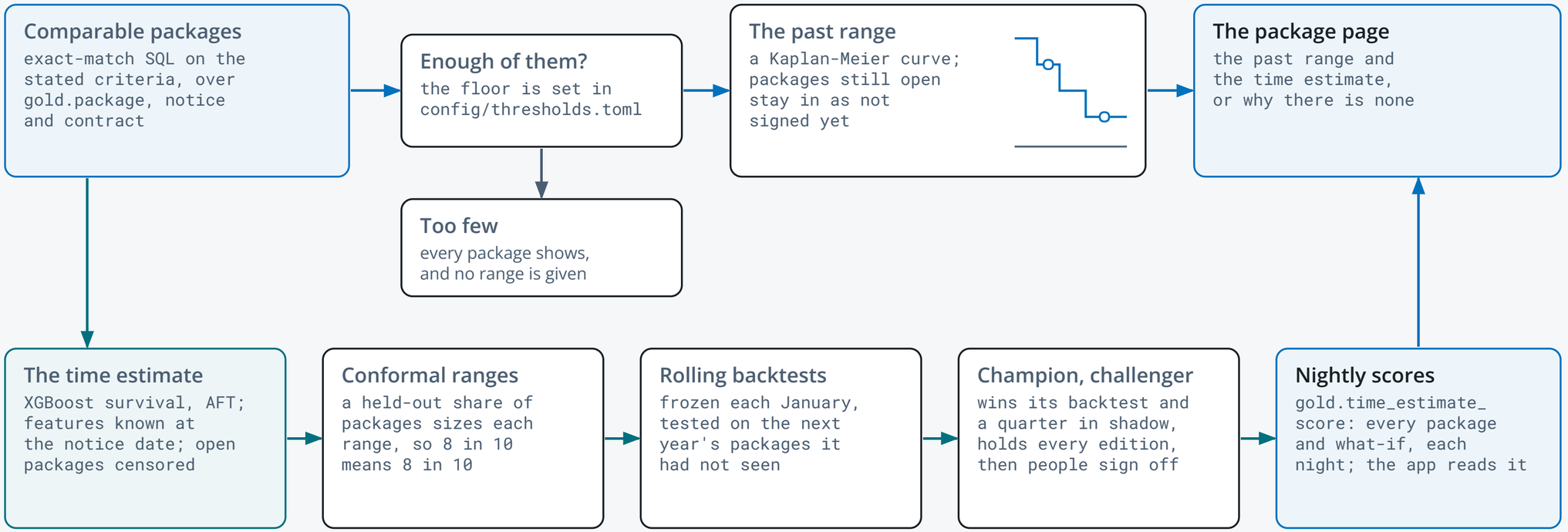
A question in one language can find evidence in another, and every quote stays in its own words



Open P127203 in Strata

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Comparable packages give the past range, and a survival model gives the time estimate



Under 20 comparable packages, Strata shows every package and gives no range. The backtest has not run on the workspace yet, so the model's track record is not shown here.

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Every sentence meets six checks before anyone sees it

Checks, in order	Sentence A cites a passage the tools returned	Sentence B has a number no tool returned
Every number matches a row a tool returned	✓ Pass	✕ Fail
Every cited ID came back in this answer's tool calls	✓ Pass	○ Not reached
Every quote is on the page it cites	✓ Pass	○ Not reached
The sentence follows from its passages the one check a model judges	✓ Pass	○ Not reached
Advice has a lesson or comparable packages behind it	✓ Pass	○ Not reached
When sources disagree, both sides show	✓ Pass	○ Not reached
Outcome	Shown	Removed, and listed under the answer with its reason

In the latest benchmark run, 27 of 43 benchmark questions passed. Strata turned down 11 of 11 questions it must refuse.

Source: Strata's benchmark run of November 1, 2026.

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When reports disagree, both stay on screen, and Strata says which it leans on



For a fact about a moment, the record made at that moment wins.

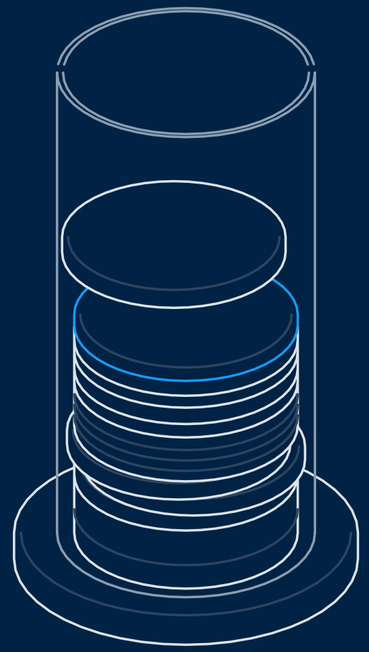
For how a project turned out, Strata leans on the most independent report written after the project closed. For a fact about a moment, the record made at that moment wins.

> In P130164, a status report rated progress toward the objective Satisfactory in August 2016. The completion report later said the mid-term review had found part of the objective compromised by major delays. Strata shows both and leans on the completion report.

Open the timeline

Source: Implementation Status and Results Report 7, P130164, August 26, 2016; Implementation Completion and Results Report, P130164, October 10, 2019, paragraph 78, page 27.

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5

How it grows

Strata learns while people decide each change, and a pilot would grow in phases.

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Three loops learn on their own schedules, and people decide each change



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The pilot grows in three phases, each measured before the next

First

Who uses Strata

World Bank procurement specialists and task team leaders in two countries

Measured before the next phase

Weekly users, drafts checked and exported, and Wrong passage reports for the sources opened

Second

Who uses Strata

Implementing agencies in the same countries

Measured before the next phase

Plan accuracy with Strata's ranges against plans made without them, and days from a new completion report to a decided lesson

Third

Who uses Strata

The public, read-only

Measured during this phase

Contract trails opened for closed projects. Active projects show the public no signals, estimates or disagreements.

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Next: more languages, more countries and more of the record

Spanish and Portuguese

Strata can already write both. A speaker still has to review each word list, French's too, and the slice needs reports in them.

More countries and sectors

The slice is one file. A new country or sector is one entry, and the same job, tools and checks apply.

Closing the data gaps

The fixes Data readiness ranks first: STEP references for contracts without one, US dollars for the gap years, older plan versions.

The whole portfolio

A higher daily page cap reads the rest of the reports, through the same page router and the same checks.

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Next: features for the people who plan and review

Amendment risk

The assessment asks for it, but no public record lists contract amendments. With STEP's records, the same survival setup could estimate it.

Answers that stream

Today the whole answer waits for every check. Next, each sentence shows the moment it passes, so long answers start sooner.

Signals that reach people

A weekly digest of new signals on the projects a person follows, sent where they already work.

The agent beside the data

When Databricks' agent tools are generally available, the orchestrator can move next to the data. The tools and checks stay.

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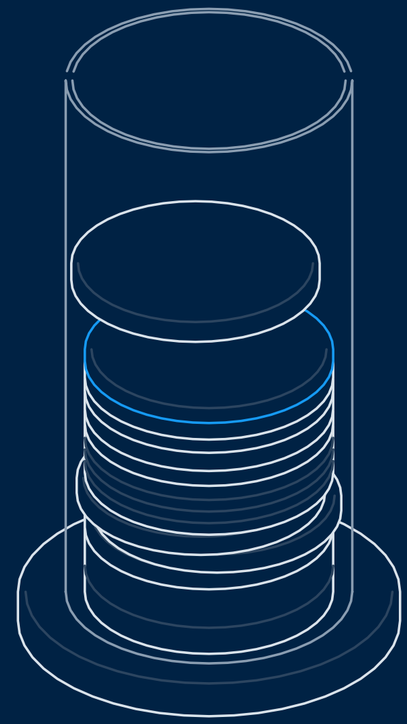
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Thank you

Chirag Lakhanpal
November 1, 2026

What would you like to ask?

Open Strata



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